

Name: _____

Class: _____

Date: _____

Homework 2.1

Binary Numbers

Score: ____ / 8

Multiple Choice

1. What is binary **1010** in decimal?

- A. 10
- B. 8
- C. 12
- D. 1010

2. What is decimal **13** in binary?

- A. 1011
- B. 1110
- C. 1101
- D. 1001

3. How many different values can be represented with **5 bits**?

- A. 5
- B. 10
- C. 25
- D. 32

4. How many bits are in one byte?

- A. 8
- B. 4
- C. 2
- D. 16

Written 1

Convert 37 to binary and 110110 to decimal.
Show your work.

Written 2

Explain overflow using 3-bit numbers as an example.

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Homework 2.2

Data Compression

Score: ____ / 8

Multiple Choice

1. What does **lossless** compression guarantee?

- A. The file becomes as small as possible
- B. Some data is permanently removed
- C. The original data can be perfectly reconstructed
- D. The file is encrypted

2. Which type of compression is best for a text document?

- A. Lossless
- B. Lossy
- C. Neither
- D. Either one works equally

3. Using run-length encoding, how is `WWWB BBBWW` compressed?

- A. W4B3W2
- B. 4W3B2W
- C. 9WB
- D. 4W5B

4. Lossy compression is commonly used for...

- A. bank records
- B. program source code
- C. streaming video
- D. medical charts

Written 1

Use RLE to compress this row of pixels: `R R R R G G B B B B B B`. How many characters are saved compared to writing each pixel letter?

Written 2

Choose lossy or lossless for each and justify: (a) a podcast for streaming, (b) a spreadsheet of grades, (c) a phone wallpaper.

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Homework 2.3

Extracting Information from Data

Score: _____ / 8

Multiple Choice

1. Which is **metadata** for a song file?

- A. The melody
- B. The lyrics
- C. The song's length and artist name
- D. The sound waves

2. Which best explains "correlation does not imply causation"?

- A. Two related variables always cause each other
- B. Two variables changing together does not prove one causes the other
- C. Data cannot show relationships
- D. Correlations are always false

3. Which is an example of **data cleaning**?

- A. Encrypting the data
- B. Deleting the whole data set
- C. Compressing the data
- D. Changing "NY" and "N.Y." to "New York" so entries match

4. **Information** is...

- A. the collection of facts and patterns extracted from data
- B. any binary number
- C. a type of hardware
- D. the same as metadata

Written 1

What personal information could a photo's metadata reveal if posted online? Why does that matter?

Written 2

Ice cream sales and drowning incidents both increase in summer. Explain why this is correlation and give an alternative explanation.

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Homework 2.4

Using Programs with Data

Score: ____ / 8

Multiple Choice

1. What does it mean to **filter** data?

- A. Arrange it in order
- B. Delete all of it
- C. Choose a subset that meets certain criteria
- D. Encrypt it

2. Which chart best shows how a value changes over time?

- A. Pie chart
- B. Single number
- C. Word cloud
- D. Line graph

3. Arranging a list of students by last name is an example of...

- A. filtering
- B. sorting
- C. compressing
- D. encrypting

4. A scatter plot is best for showing...

- A. parts of a whole
- B. alphabetical order
- C. one single value
- D. the relationship between two variables

Written 1

A menu data set has: dish, price, calories, isVegan. List the steps to find the 3 cheapest vegan dishes.

Written 2

Choose a visualization and explain: (a) how the number of app users grew each month this year; (b) whether hours of sleep relate to test scores.
